COUNTEREXAMPLES TO AN ASSOCIATED-PRIME CONJECTURE OF FOULI, HÀ, AND MOREY

Abstract. We refute the only-if implication in Conjecture 3.7 of Fouli, Hà, and Morey on associated primes obtained after adjoining separated stars to an edge ideal. A single star on the four-cycle already gives a counterexample, for which we determine the complete set of associated primes. More generally, over every field and for every $q \geq 1$, the cycle C_{3q+1} admits q pairwise separated stars such that the homogeneous maximal ideal is an embedded associated prime after adjoining the stars, but is not obtained from any minimal prime of the edge ideal in the conjectured manner.

1. A four-vertex counterexample

Let $I(G)$ denote the edge ideal of a finite simple graph G. Fouli, Hà, and Morey conjectured that if $I = I(G)$ and $f_1, \dots, f_q$ are separated stars on I, then

$$P \in \operatorname{Ass}(R/(I, f_1, \dots, f_q)) \iff P = (Q, f_1, \dots, f_q) \tag{1}$$

for some $Q \in \operatorname{Ass}(R/I) = \operatorname{Min}(R/I)$; see [2, Definitions 2.1–2.2 and Conjecture 3.7]. The following example disproves the only-if implication in (1) with one star.

Theorem 1. *Let K be any field and set*

$$R = K[x_1, x_2, x_3, x_4], \qquad I = (x_1x_2, x_1x_3, x_2x_4, x_3x_4).$$

Let

$$f = x_1 + x_2 + x_3, \qquad J = (I, f), \qquad \mathfrak{m} = (x_1, x_2, x_3, x_4).$$

Then f is a star on I, centered at x_1, and

$$(J : x_2 + x_3) = \mathfrak{m}.$$

Moreover,

$$\operatorname{Ass}(R/J) = \{(x_1, x_2, x_3),\ (x_1, x_4, x_2 + x_3),\ \mathfrak{m}\}.$$

The first two primes are of the form predicted by (1), whereas the embedded prime $\mathfrak{m}$ is not. Consequently, the only-if implication in Conjecture 3.7 of Fouli, Hà, and Morey is false.

Proof. The generators of I divisible by x_1 are x_1x_2 and x_1x_3, so f is a star centered at x_1. Separation is vacuous for a single star.

Eliminate x_1 using $f = 0$ and put

$$u = x_2 + x_3, \qquad v = x_3, \qquad w = x_4.$$

Key words and phrases. edge ideal, associated prime, separated stars, cycle.

Then

$$R/J \cong K[u,v,w]/L, \qquad L = (u^2, uv, uw, vw).$$

Since $u \notin L$ and

$$(L : u) = (u, v, w),$$

the class of u has annihilator (u,v,w). Because $f \in J$, the preimage of this colon ideal in R is $(J : x_2 + x_3)$. Hence $(J : x_2 + x_3) = \mathfrak{m}$ and $\mathfrak{m} \in \operatorname{Ass}(R/J)$.

The irredundant primary decomposition

$$L = (u,v) \cap (u,w) \cap (u^2, v, w)$$

yields

$$\operatorname{Ass}(K[u,v,w]/L) = \{(u,v), (u,w), (u,v,w)\}.$$

Pulling these primes back to R gives the displayed set. Finally,

$$\operatorname{Min}(R/I) = \{(x_1, x_4), (x_2, x_3)\},$$

and adjoining f to these two primes gives, respectively, $(x_1, x_4, x_2 + x_3)$ and (x_1, x_2, x_3). Thus $\mathfrak{m}$ is the additional embedded associated prime. □

2. Counterexamples with arbitrarily many stars

Theorem 2. *Fix $n \geq 1$ and a field K. Let $G_n = C_{3n+1}$ have vertices*

$$a_1, c_1, b_1, \ldots, a_n, c_n, b_n, z$$

in cyclic order

$$a_1 - c_1 - b_1 - a_2 - c_2 - b_2 - \cdots - a_n - c_n - b_n - z - a_1.$$

Set

$$R = K[a_1, c_1, b_1, \ldots, a_n, c_n, b_n, z], \qquad I = I(G_n),$$

and let

$$f_i = a_i + c_i + b_i \quad (1 \leq i \leq n), \qquad J = (I, f_1, \ldots, f_n).$$

Then $f_1, \ldots, f_n$ are pairwise separated stars on I. If

$$U = \prod_{i=1}^{n} (a_i + b_i), \qquad \mathfrak{m} = (a_1, c_1, b_1, \ldots, a_n, c_n, b_n, z),$$

then

$$(J : U) = \mathfrak{m}.$$

Consequently, $\mathfrak{m}$ is an embedded associated prime of R/J, but

$$\mathfrak{m} \neq (Q, f_1, \ldots, f_n) \qquad \textit{for every } Q \in \operatorname{Ass}(R/I) = \operatorname{Min}(R/I).$$

Proof. The support of f_i is the closed neighborhood of c_i. Thus f_i is a star centered at c_i. The supports are pairwise disjoint, and no center is adjacent to a noncenter variable in another support, so the stars are pairwise separated.

Write $u_i = a_i + b_i$, so $U = u_1 \cdots u_n$. For each i,

$$a_i u_i = a_i f_i - a_i c_i \in J, \qquad b_i u_i = b_i f_i - b_i c_i \in J,$$

and

$$c_i u_i = c_i a_i + c_i b_i \in I.$$

Multiplying by $\prod_{j \neq i} u_j$ shows that every a_i, b_i, c_i belongs to $(J : U)$.

We also have $zU \in I$. Indeed, every monomial term of zU has the form

$$M = z\epsilon_1 \cdots \epsilon_n, \qquad \epsilon_i \in \{a_i, b_i\}.$$

If $\epsilon_1 = a_1$, then $za_1 \mid M$. Otherwise $\epsilon_1 = b_1$. Unless some edge $b_i a_{i+1}$ with $1 \leq i < n$ divides M, induction gives $\epsilon_i = b_i$ for every i, and then $b_n z \mid M$. Thus every monomial term of zU lies in the monomial ideal I. It follows that

$$\mathfrak{m} \subseteq (J : U).$$

To prove that the colon is proper, let

$$A = K[t_1, \ldots, t_n]/(t_1^2, \ldots, t_n^2)$$

and define a K-algebra homomorphism $\varphi : R \to A$ by

$$\varphi(a_i) = t_i, \qquad \varphi(c_i) = -t_i, \qquad \varphi(b_i) = 0, \qquad \varphi(z) = 0.$$

Every edge generator of I and every f_i maps to zero, while

$$\varphi(U) = t_1 \cdots t_n \neq 0,$$

since the squarefree monomials form a K-basis of A. Hence $U \notin J$. Since $(J : U)$ is proper and contains the maximal ideal $\mathfrak{m}$, we obtain $(J : U) = \mathfrak{m}$. Therefore $\mathfrak{m} \in \operatorname{Ass}(R/J)$. Moreover, the prime ideal

$$H = (a_1, c_1, b_1, \ldots, a_n, c_n, b_n)$$

satisfies $J \subseteq H \subsetneq \mathfrak{m}$, so $\mathfrak{m}$ is embedded.

It remains to show that $\mathfrak{m}$ is not of the form in (1). Let $Q \in \operatorname{Min}(R/I)$. The variables outside Q form a maximal independent set S of G_n, hence a dominating set. Each closed neighborhood in a cycle has three vertices, so

$$3|S| \geq 3n + 1, \qquad \text{and therefore} \qquad |S| \geq n + 1.$$

Modulo Q, the degree-one component of $\mathfrak{m}/Q$ has dimension $|S|$, whereas the images of $f_1, \ldots, f_n$ span a space of dimension at most n. Hence $(Q, f_1, \ldots, f_n) \neq \mathfrak{m}$. Finally, I is squarefree, so $\operatorname{Ass}(R/I) = \operatorname{Min}(R/I)$. □

Remark 3. Up to relabeling, the linear forms in Theorem 2 were recorded by Tran as an initially regular sequence realizing the depth of C_{3n+1}; see [3, Remark 3.2] and the general construction in [1, Theorem 3.11]. The contribution here is the explicit annihilator $(J : U) = \mathfrak{m}$ and the resulting counterexamples to Conjecture 3.7. The reverse implication in (1) is not addressed.

References

[1] L. Fouli, T. H. Hà, and S. Morey, *Initially regular sequences and depths of ideals*, J. Algebra **559** (2020), 33–57.

[2] L. Fouli, T. H. Hà, and S. Morey, *Regular sequences of linear forms on monomial ideals*, arXiv:2606.15125v1 [math.AC], 2026.

[3] L. Tran, *Initially regular sequences on cycles and depth of unicyclic graphs*, arXiv:2309.07286v1 [math.AC], 2023.